Alberto Reyes Castro

Applied AI Engineer | Multimodal & On-Device ML | MSc AI Technology

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Portfolio | GitHub | LinkedIn | Kaggle

UK Student visa valid to May 2028: up to 20 hours/week during term time; full-time during official university vacations.

Professional Summary

Physics graduate and MSc Artificial Intelligence Technology student at Northumbria University London, currently performing at Distinction level. Applied AI engineer focused on multimodal retrieval, efficient on-device inference, model evaluation and reliable AI systems. I build systems end to end, from model selection and benchmarking through implementation, physical-device validation and human-review workflows, and I am seeking Summer 2027 machine-learning / Applied AI internships.

Technical Skills

- **Machine learning & evaluation:** Python, PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, TensorFlow/Keras (coursework), pandas, model benchmarking, imbalanced learning, cross-validation, threshold analysis, Hit@k / retrieval evaluation, error analysis.
- **Multimodal, mobile & local AI:** vision-language models, TinyCLIP, ONNX Runtime, INT8 quantisation, Android, React Native/Expo, Kotlin, SQLite, latency/memory profiling, faster-whisper, FFmpeg, Gemini API, Ollama.
- **Engineering & scientific computing:** TypeScript, React, Next.js, Git/GitHub, Linux, Bash, PowerShell, SQL, REST APIs, Docker fundamentals, NumPy, SciPy, Matplotlib, Jupyter, LaTeX/Overleaf.

Relevant Experience

Applied AI Engineer - On-Device Multimodal ML **Jul 2026-Present**

TeachScribe - Technical collaboration, London / Hybrid | Full-time Jul-Oct 2026; part-time alongside MSc since Oct 2026

- Designed, built and evaluated a fully on-device multimodal photo-retrieval prototype for Android for TeachScribe, a UK Early Years platform in use in 50+ schools; local photos are ranked against written observations with a person confirming every match.
- Benchmarked five pretrained vision-language models on curated retrieval evaluations covering Hit@1/Hit@5, model size, CPU cost, mobile deployability and checkpoint licensing; selected TinyCLIP 39M/16 (17/20 Hit@1, 18/20 Hit@5 on a 20-query desktop set with 16 candidate images).
- Implemented INT8 ONNX Runtime inference with native Kotlin preprocessing, reusable sessions and a persistent SQLite embedding cache; dynamic INT8 quantisation reduced the text encoder from approximately 170 MiB to 43 MiB (74.8%).
- On a Huawei MediaPad M3 Lite 8 (Android 7), 108/108 cached searches over 44 photos completed at 441 ms median / 459 ms P95; lazy image sessions held idle memory at 76.3 MiB PSS versus 166.6 MiB with preloading. Source code and internal product materials remain private.

Python & Scientific Communication Intern **Mar 2025-May 2025**

Instituto de Astrofisica de Canarias (IAC), Tenerife | 250h | Final grade: 9.2/10

- Developed Python workflows for multilingual interview transcription/subtitling and semi-automated matching of 67 outreach articles against 700+ scientific publications using text preprocessing, regex and fuzzy matching with manual review for uncertain cases.
- Produced technical documentation and science communication in English and Spanish, including scientific interview material and published IAC outreach content.

Selected Projects

ModelHawk - ML Notebook Audit Prototype - GitHub

Next.js/TypeScript tool that parses classification notebooks and applies deterministic and heuristic checks for leakage risk, metric choice, class imbalance, validation quality and reproducibility. Optional examiner notes use the Gemini API with a local Llama 3.2 fallback through Ollama; reports export to LaTeX/PDF.

Agent Control Center - Multi-Agent Orchestration - GitHub | Demo

Public interactive simulation (six failure scenarios) of the control plane behind a private agent-orchestration harness, illustrating capability-aware routing, execution leases, bounded recovery, separate review, deterministic checks and a human merge gate. Authored demo data; no live telemetry or provider calls.

Credit Card Fraud Detection - GitHub

MSc machine-learning project (85/100) on 284,807 transactions with leakage-controlled preprocessing, six classifiers, imbalance strategies, stratified cross-validation and validation-based thresholding. Tuned XGBoost: 0.8827 average precision, 91.2% precision and 84.7% recall.

Adaptive Whisper Transcription Pipeline - GitHub

Reusable faster-whisper workflow with CPU/GPU-aware model selection, VAD, repetition safeguards, caching, quality checks and TXT/SRT export; evolved from scientific interview transcription work at the IAC.

AIoT Smart Home Prototype - GitHub

ESP32-S3 / Edge Impulse / Wokwi system with profile-aware automation and deterministic safety fallbacks; simulated, rule-labelled held-out set: 85.9% accuracy, 0.99 AUC and 0.89 weighted F1.

IAC Publication-Press Note Matching - GitHub

Semi-automated Python workflow comparing 67 outreach articles with 700+ scientific publications using text preprocessing, regex and fuzzy matching with manual review for uncertain matches.

Computational Physics - Numerical Methods - GitHub

Scientific Python notebooks covering interpolation, numerical integration, ODE solvers and stochastic simulation, with reproducible derivations and validation against analytical or reference behaviour.

Open-Source Contribution

Vizro / Dash Rendering Regression Analysis **2026**

Reproduced and narrowed a Vizro/Dash tab-rendering regression by comparing dependency versions, reproducing the behaviour independently in pure Dash and separating framework behaviour from Plotly graph rendering. The analysis was acknowledged by a Vizro maintainer and helped narrow the source of the regression. Maintainer comment.

Selected Publications & Scientific Communication

- *The Partial Solar Eclipse of March 2025* - authored Spanish-language science-outreach article for the IAC Via Lactea blog during the 2025 internship, explaining the observing circumstances and public-science significance of the March 2025 partial solar eclipse.
- *The WEAVE spectrograph captures the interaction of supernova SN 2023ixf with its surroundings one year after its explosion* - wrote the English-language IAC News version communicating current astrophysical research for an international audience, translating specialist results into accurate public-facing scientific prose.
- *El eclipse parcial que recordo a Canarias el valor de mirar al cielo* - co-authored feature published in *Astronomia* magazine, produced during the IAC internship and focused on the scientific and public-engagement significance of the 29 March 2025 eclipse in the Canary Islands.

Education

MSc Artificial Intelligence Technology with Advanced Practice **Jan 2026-Jan 2028**

Northumbria University, London Campus - London, UK

Currently performing at Distinction level, with first-semester marks including **85/100** (end-to-end machine-learning project on credit-card fraud detection), **75/100** and **70/100**. Relevant study includes Programming for AI, Machine Learning on Cloud, Artificial Intelligence for IoT, Information Governance & Cyber Security, Contemporary Computing and Digital Technologies, and Advanced Practice.

BSc Physics **Sep 2018-Jul 2025**

Universidad de La Laguna - Tenerife, Spain | Final grade: 7.13/10

Final Degree Project written and defended in English on structural mechanisms of ferroelectricity and ferroelasticity in beta-prime rare-earth molybdates, **9.4/10**. Academic Merit Convocation Award (2025). Selected high marks include General Relativity 10/10, Acquisition and Transmission of Signals and Images 9.9/10, Differential Calculus 9.5/10 and Electronic Physics 9.0/10.

Languages & Selected Training

- **Languages:** Spanish - Native; English - C1 Advanced (IELTS Academic UKVI 7.0; Cambridge Advanced C1); Japanese - intermediate study.
- **Artificial Intelligence and Big Data Applied to the Healthcare Sector** - 325 hours, Docencia y Multimedia Canarias for the Canary Islands Employment Service, Sep-Dec 2025: Python, machine learning, Big Data systems, and probability/statistics with R.
- **AWS Educate Machine Learning Foundations** (2026); **Google Cloud Skills Boost** selected AI/data courses (2025); **NASA Open Science 101** (2025); **International Open Innovation Training (IDEA-UP)**, Wroclaw University of Science and Technology (2023).
- **English certifications:** IELTS Academic UKVI 7.0/9.0 (British Council, 2025) and Cambridge Advanced Certificate C1 (2022).
- **Recommendations:** signed academic recommendation from a Full Professor of Applied Physics and professional recommendation from the IAC internship supervisor, both available through the portfolio.

Portfolio: albertoreyescastro.github.io